

Design of a 5.4 kWp Grid-connected Residential Rooftop Solar PV system

(Complete Report)

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Location: Rajshahi, Bangladesh
System type: Grid-connected Solar system
Inverter Capacity: 5 kWp AC
Battery: LiFePO₄ 4.8 kWh nominal, 48 V
Tools used: nanoCAD v5, ETAP

2.1 Executive Summary

The proposed system consists of 12×450 W solar PV modules, providing a total installed DC capacity of 5.40 kWp. The PV array is configured as two independent series strings, with six modules connected in series per string. Each string has a nominal capacity of 2.70 kWp and is connected to a separate MPPT input of a 5.0 kW GoodWe GW5000-ES-C10 hybrid inverter (a generic choice).

The system also includes a 48 V, 100 Ah LiFePO₄ battery with a nominal energy capacity of 4.8 kWh. The battery is intended primarily for backup of essential household loads rather than the entire residential load.

The project covers:

- Electrical load assessment
- Solar PV sizing
- Battery sizing
- Inverter selection
- PV string configuration
- Cable sizing
- Protection planning
- Rooftop layout
- Electrical equipment layout
- Single-line diagram
- Equipment schedule/BOM
- Engineering documentation

The project is intended as a conceptual engineering portfolio project. Actual site measurements, structural certification, utility approval, detailed protection coordination, equipment procurement, installation, testing, and commissioning are excluded here.

2.2 Project Objectives

The main objectives of the project were:

1. Note down the assumed residential electrical load and daily energy consumption.
2. To determine a suitable PV array capacity.
3. Develop a practical PV string configuration.
4. To select a compatible hybrid solar inverter & battery capacity for essential-load backup.
5. Perform preliminary DC and AC cable sizing.
6. To identify appropriate electrical protection devices.
7. Overall learn and demonstrate engineering drawing skills (2D) using CAD softwares.
8. Generate a detailed SLD of overall system.
9. Organize the design information into a professional engineering report.

2.3 Design Basis and Assumptions

Parameter	Design Basis
Location	Rajshahi, Bangladesh
Roof type	Flat RCC rooftop
Assumed roof size	10 m × 8 m
Roof area	80 m ²
Orientation	South-facing
Shading	Low/minimal shading assumed
Grid supply	230 V, single-phase
Frequency	50 Hz
PV module	JA Solar 450 W (12 modules)
PV configuration	6S + 6S strings
Hybrid inverter	GoodWe GW5000-ES-C10 (5 kW AC out)
Battery	48 V, 100 Ah LiFePO ₄
Battery nominal energy	4.8 kWh
Assumed battery DoD	80%
Average HSP	4.88 h/day

2.4 Residential Load Assessment

An assumed residential load profile was developed to estimate the daily electricity requirement.

Load	Quantity	Rating	Time (h/day)	Daily Energy
LED lighting	12	20 W	6 h	1.44 kWh
Ceiling fans	6	80 W	8 h	3.84 kWh
Refrigerator	1	400 W	10 h	4.00 kWh
Television	2	120 W	5 h	1.20 kWh
Computers	2	150 W	4 h	1.20 kWh
Washing machine	1	750 W	1 h	0.75 kWh
Water pump	1	750 W	1 h	0.75 kWh
Air conditioners	2	1200 W	4 h	9.60 kWh
Kitchen Utilities	-	1000 W	3 h	3.00 kWh
Chargers	6	100 W	4 h	2.40 kWh
Miscellaneous	-	200 W	1 h	0.20 kWh
Total				28.38 kWh/day

This is the estimated energy requirement of our conceptual client's house. Now to find out the approx. power demand of the house, I followed BNBC (Bangladesh National Building Code) standards to calculate diversity in power demand –

Load	Category	Connected Load	Demand Factor	Power Demand
Lights, Fans	Essentials	0.72 kW	0.66	0.48 kW
TV, PC, Chargers	General	1.34 kW	0.40	0.54 kW
Refrigerator	Power	0.4 kW	1.00	0.40 kW
Air Conditioner	Power	2.4 kW	1.00	2.40 kW
Water Pump	Power	0.75 kW	0.50	0.38 kW
Washing machine	Power	0.75 kW	0.50	0.38 kW
Kitchen Utilities	General	1 kW	0.40	0.40 kW
Total				4.96 kW

2.5 PV System Sizing

The selected PV module is a 450 W JA Solar module. The total PV capacity is:

$$12 \times 450 \text{ W} = 5.4 \text{ kWp}$$

An assumed average peak-sun-hours value of 4.88 hours/day and a performance ratio of 0.80 were used for preliminary energy estimation. According to such, estimated daily PV energy generation:

$$\text{Energy} = \text{PV capacity} \times \text{HSP} \times \text{PR} = 5.40 \times 4.88 \times 0.80 \approx 20.73 \text{ kWh/day}$$

20.7 kWh energy harvesting per day is a good amount, considering daily energy demand is 28.4 kWh. This means the PV system can cover for 72% of daily need. The rest can be drawn from grid line. This is an estimated figure of energy generated by PV in 5-hour duration. However, it will be more than that in clear sunny days when sunlight is available almost 7-8 hours/day. Estimated monthly energy generation:

$$20.73 \text{ kWh/day} \times 30 \text{ days} \approx 622 \text{ kWh/month}$$

Say that, during peak-sun-hours, power demand of the house is 3.2 kW. Then energy required:

$$3.2 \text{ kW} \times 4.88 \text{ h} \approx 15.61 \text{ kWh}$$

So, excess energy remained:

$$20.73 - 15.61 = 5.12 \text{ kWh}$$

Which can be sent back to grid + use for battery charging (if required).

These values are engineering estimates rather than a guaranteed generation figure. Actual generation will depend on irradiation, temperature, shading, module orientation, soiling, system losses, inverter efficiency, weather and operating conditions.

2.6 PV String Configuration

The PV array was intentionally configured as two independent six-module series strings.

String 1: $6 \times 450 \text{ W} = 2.70 \text{ kWp}$ Connected to: **MPPT 1**

String 2: $6 \times 450 \text{ W} = 2.70 \text{ kWp}$ Connected to: **MPPT 2**

$$\text{Total} = 2.70 + 2.70 = 5.40 \text{ kWp}$$

The selected module has approximated electrical characteristics of:

- ✓ $V_{mp} = 32.82 \text{ V}$
- ✓ $V_{oc} = 39.30 \text{ V}$
- ✓ $I_{mp} = 13.71 \text{ A}$
- ✓ $I_{sc} = 14.48 \text{ A}$

For each series string:

$$V_{mp} \approx 6 \times 32.82 = 196.92 \text{ V}$$

$$V_{oc} \approx 6 \times 39.30 = 235.80 \text{ V}$$

This means, primarily upon load connected scenario, max voltage from each string we can get is 196 V. As two strings actually becomes parallel inside the inverter, it adds up current and power while maintain appropriate voltage intelligently.

[Note: OC voltage rating of each module is specified as a rating at 25 deg. C temp. and temperature coefficient is -0.250%/C (*minus sign because above 25°C, voltage starts to drop*). So, say current temp. is 10°C, then temp. difference = 15, performance increased by = $15 \times 0.250\% = 3.75\%$. So finally, OC voltage at 10°C = $39.3 \times 1.0375 = 40.77 \text{ V}$]

2.7 Inverter Selection

The selected hybrid inverter is:

GoodWe GW5000-ES-C10

Key design characteristics considered include:

- 5.0 kW nominal AC output
- 230 V single-phase operation
- 50 Hz operation
- Two MPPT inputs
- 48 V battery architecture
- Suitable PV voltage/current operating range
- Hybrid operation with battery storage

The inverter was selected because its electrical input characteristics are compatible with the proposed six-module dual series strings.

The approximate six-module operating voltage of 196.92 V falls within the inverter's published MPPT operating range (60 V – 550 V). The PV current of approximately 13.71 A per string is also below the inverter's specified maximum input current per MPPT (20 A, I_{SC} max. is 26 A).

2.8 PV-Inverter Compatibility Check

Parameter	Design Value	Inverter Limit	Result
Total PV capacity	5.40 kWp	10 kW max PV	Pass
String V_{mp}	196.92 V	60–550 V MPPT range	Pass
String V_{oc}	235.80 V	600 V max PV voltage	Pass
String I_{mp}	13.71 A	20 A/MPPT	Pass
String I_{sc}	14.48 A	26 A/MPPT	Pass
Number of strings	2	2 MPPTs	Pass
Strings per MPPT	1	1 string/MPPT	Pass

A preliminary cold-temperature V_{oc} check was also considered.

Assuming a minimum module temperature of approximately 10°C and a V_{oc} temperature coefficient of - 0.250%/°C:

Corrected per module $V_{oc} = 40.77$ V

Therefore:

Corrected per string $V_{oc} = 244.6$ V

This also remains well below the inverter's maximum PV voltage.

2.9 Battery Sizing

A conceptual battery capacity of:

48 V $\times$ 100 Ah = 4.8 kWh

was selected. Assuming 80% usable depth of discharge:

Usable DC energy = $4.8 \times 0.80 = 3.84$ kWh

Using an assumed 90% battery/conversion efficiency:

Usable AC energy $\approx 3.84 \times 0.90 \approx 3.46$ kWh (actual usable energy)

So, if we can limit the electricity usage during power shortages at essentials only loads (say 1 kW load) then the battery can provide power supply for more than 3h. Which is enough time for urban areas though.

2.10 Essential Backup Load

The battery was not designed to supply the complete household load rather than as a backup for essential loads. The selected essential loads are:

Essential Load	Approx. Power
8 LED lights	160 W
4 fans	320 W
Refrigerator	400 W
1 TV / 1 Computer	150 W
Chargers, Wi-Fi	200 W
Total	1230 W

Therefore:

Essential backup load ≈ 1.2 kW

Estimated backup duration:

Backup time = $3.46 \text{ kWh} / 1.2 \text{ kW} \approx 3$ hours

This estimate assumes the battery starts at its usable energy level and excludes additional losses or changes in load. High-power loads such as air conditioners, the water pump and washing machine were excluded from the essential backup circuit.

2.11 DC Cable Sizing

A preliminary **4 mm² copper PV cable** was considered.

For an assumed 25 m one-way distance:

Total loop length = 50 m

Using copper resistivity of approximately $0.0175 \Omega \cdot \text{mm}^2/\text{m}$ and $I_{\text{mp}} = 13.7 \text{ A}$

Voltage drop $\approx 3.00 \text{ V}$ (Calculated using $[V = (2 \cdot I \cdot L \cdot \rho) / A]$ formula)

Percentage voltage drop:

$(3/196.9) \times 100\% = 1.52\%$

This is considered acceptable for the preliminary conceptual design.

Actual cable length and final cable size should be determined from the final installation route, environmental conditions, allowable ampacity, voltage-drop requirements and manufacturer recommendations.

2.12 AC Cable Sizing

The approximate inverter output current is:

$I = P/V = 5000/230 \approx 21.74 \text{ A}$

A preliminary:

6 mm² copper AC cable was considered.

For an assumed 20 m one-way route:

Total loop length = 40 m

Using copper resistivity of approximately $0.0175 \Omega \cdot \text{mm}^2/\text{m}$ and $I = 21.74 \text{ A}$:

Voltage drop $\approx 2.54 \text{ V}$

Percentage voltage drop:

$(2.54/230) \times 100\% \approx 1.10\%$

A preliminary: **32 A, 2-pole MCB** was selected for the conceptual design.

2.13 Electrical Protection

The conceptual system includes the following protection arrangements:

DC side

- DC isolation
- DC surge protection as required
- Appropriate PV string protection based on the final equipment arrangement
- Battery DC isolation/protection

AC side

- AC isolation
- Type II AC SPD
- 32 A, 2-pole MCB as a preliminary selection
- Main distribution board protection
- Backup distribution board protection

Earthing and bonding

The design provides for bonding of:

- PV module frames
- PV mounting structure
- Inverter PE
- Electrical equipment
- Distribution-board protective earth

A preliminary 6 mm² copper sheet conductor was considered, subject to final design verification. No assumed earth resistance value has been used because actual site measurements were not available.

2.14 Rooftop PV Layout

The rooftop layout was developed in nanoCAD v5.

The assumed rooftop dimensions are:

10,000 mm × 8,000 mm

The layout includes:

- Roof boundary
- Parapet/clearance zone
- Staircase enclosure
- PV modules
- Maintenance walkway
- Inverter location
- Equipment labels
- Dimensions
- North direction
- Engineering notes

The 12 modules are arranged as two separate six-module strings corresponding to the electrical design.

Drawing: RoofTop_Layout.dwg

Drawing No.: PV-01

2.15 Electrical Physical Layout

A separate electrical layout was developed in nanoCAD to represent the physical arrangement of major electrical equipment. Roof dimension was not considered here as it is an electrical component layout drawing.

The layout includes:

- Hybrid inverter
- Battery
- DCDB box
- ACDB box
- Main distribution board
- Net meter
- Grid line
- Cable routes
- Earthing/bonding
- Equipment tags
- Dimensions and annotations

Drawing: Electrical_Layout.dwg

Drawing No.: PV-02

The physical layout is intended to show equipment arrangement and routing rather than detailed construction or installation drawings.

2.16 Single-Line Diagram

The electrical single-line diagram was developed in ETAP software.

The SLD represents the electrical flow from the PV array through the hybrid inverter and protection equipment to the battery, residential loads and utility grid.

The major electrical paths are:

PV path

PV String 1 → DC protection/isolation → MPPT 1 → Hybrid inverter

PV String 2 → DC protection/isolation → MPPT 2 → Hybrid inverter

Battery path

48 V battery → Battery protection/isolation → Hybrid inverter

AC path

Hybrid inverter → AC isolation → AC SPD → 32 A 2P MCB → Main DB

Load path

Main DB → Backup DB → Essential loads

Grid path

Grid → Net meter → Main electrical system

The SLD also identifies the protective-earth/bonding arrangement, major equipment tags & naming abbreviations.

Drawing: 5.4kW_Residential_PV_SLD.oti

Drawing No.: PV-03

2.17 Equipment Schedule / BOM

The major equipment identified for the project is summarized below.

TAG	EQUIPMENT	MANUFACTURER	RATING	QTY
PV-01	PV modules	JA Solar JAM54S40	450 W	12
INV-01	Hybrid inverter	GoodWe GW5000-ES	5.0 kW	1
BAT-01	LiFePO ₄ battery	Generic	48 V, 100 Ah	1
DC-01	DCDB box	Suntree	2-In/2-Out, 600V	1
DC-02	DC CB	Beny Electric	20 A, 500 V	1

TAG	EQUIPMENT	MANUFACTURER	RATING	QTY
BAT-02	Battery protection	To be selected	Final rating required	1
AC-01	AC isolator	To be selected	Final rating required	1
AC-02	AC CB	To be selected	Final rating required	1
AC-03	ACDB box	Havells	32 A, 2P preliminary	1
AC-04	Bi-directional Net meter	Secure Meters	230 V, 30 A/60 A, Class I	1
ELC-01	Surge Arrester	Ingesco	60 kA, 40m coverage	1

Final manufacturers and models for protection devices and battery equipment should be selected based on actual procurement requirements and equipment compatibility.

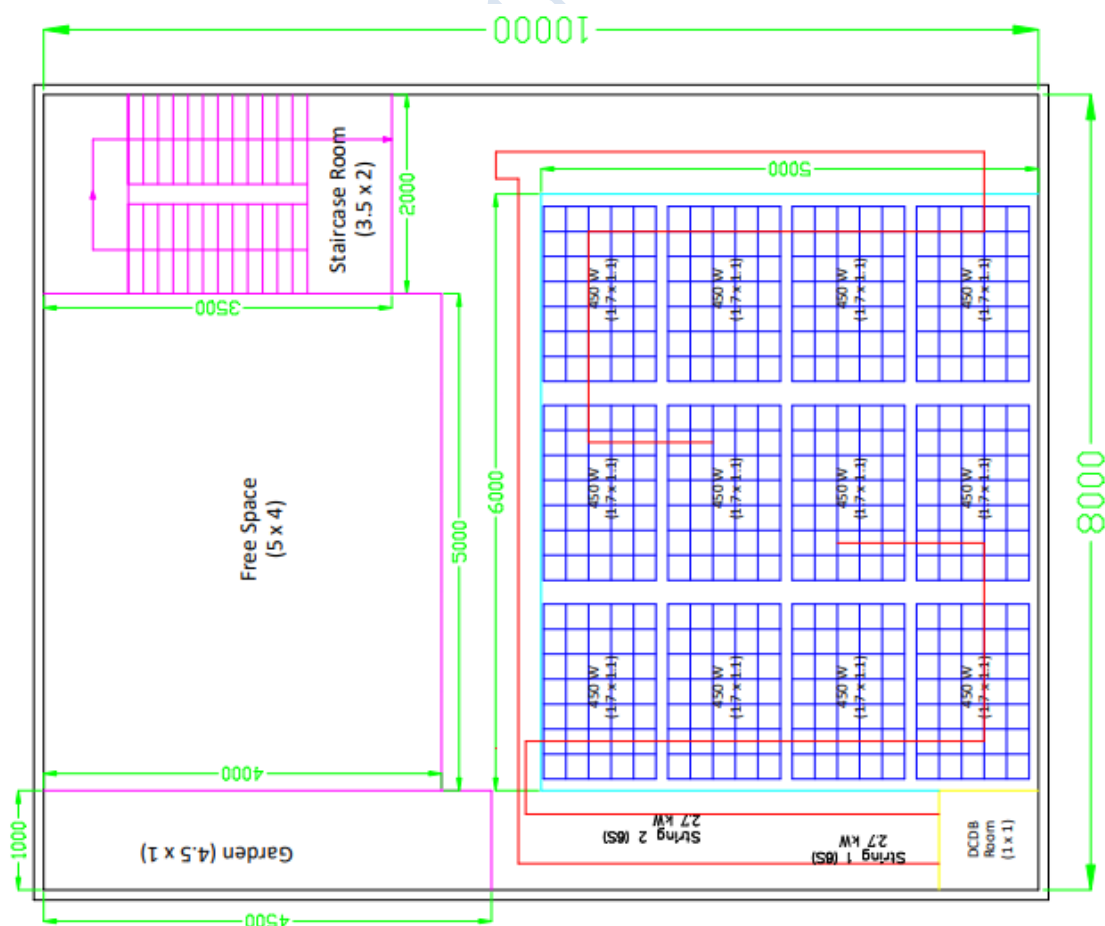
2.18 Project Deliverables

The completed project consists of:

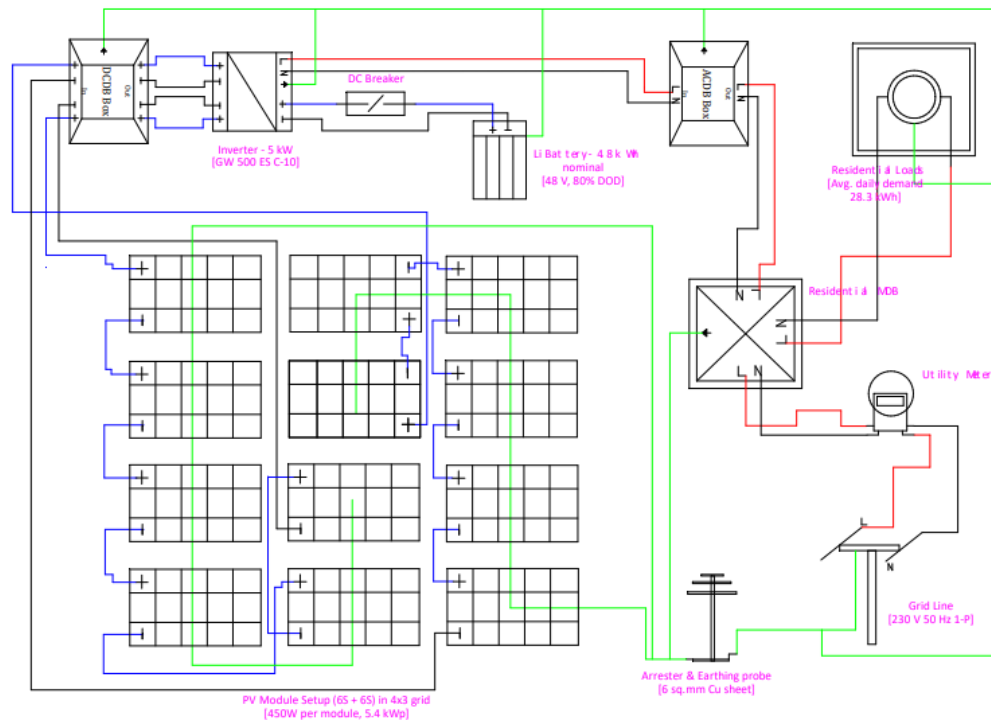
1. Site assessment and design assumptions
2. PV sizing and load calculation
3. Cable sizing and protection calculation
4. Rooftop PV layout
5. Electrical physical layout
6. Electrical single-line diagram
7. Equipment/BOM schedule
8. Final project report

The project demonstrates a workflow from preliminary design information through engineering calculations and finally to technical drawings and documentation. Here are screenshots of drawings:

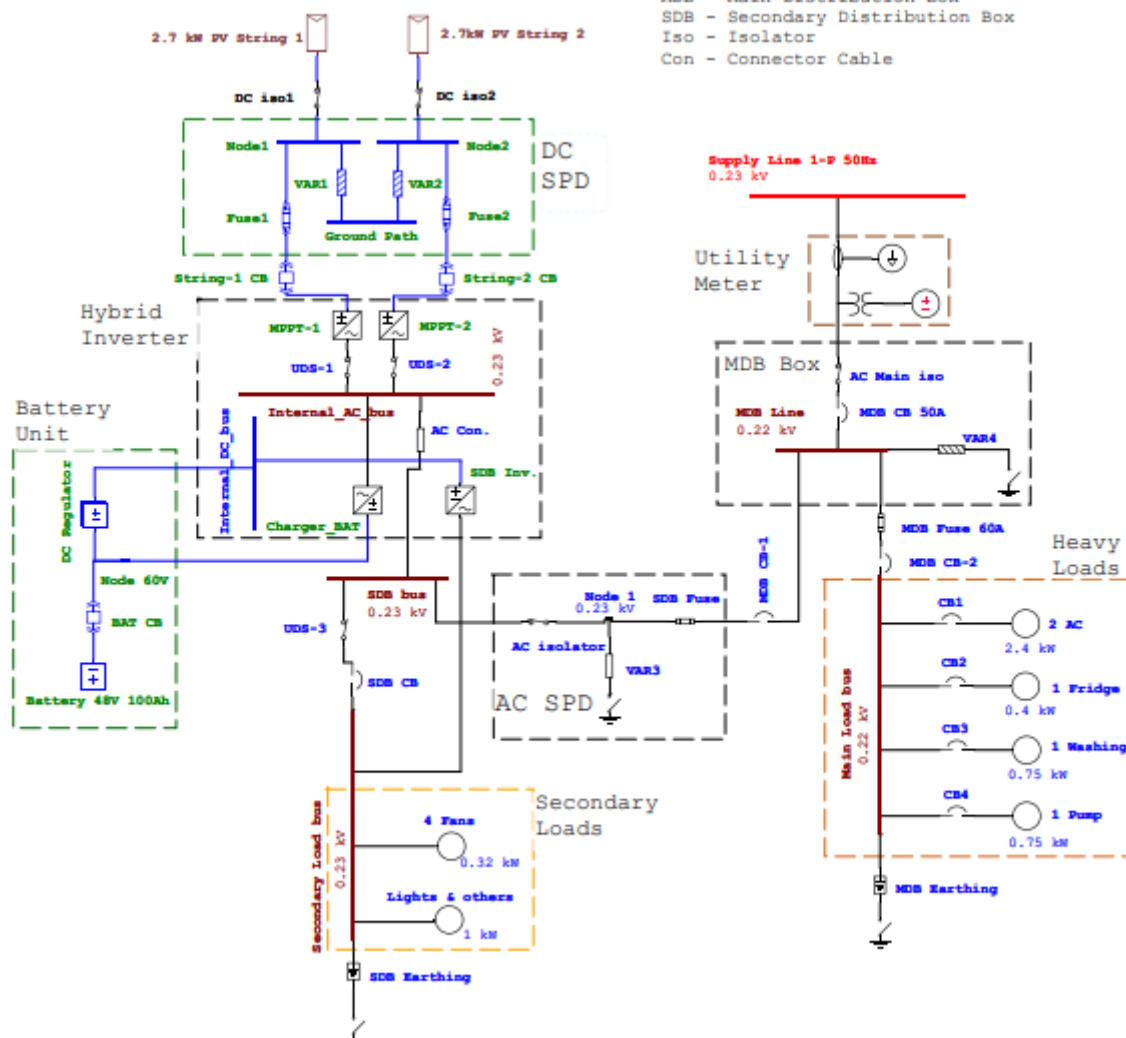
RoofTop_Layout:



Electrical_Layout:



PV_System_SLD:



2.19 Engineering Considerations and Limitations

This project is a conceptual portfolio design and therefore has several limitations.

Site limitations

Actual site measurements were not available. Roof dimensions, obstacles, shading and orientation were assumed.

Structural limitations

No structural analysis, wind-load calculation, RCC assessment, mounting-anchor calculation or roof-load certification was performed.

Solar resource limitations

The solar-generation estimate uses an assumed average peak-sun-hours value and performance ratio. Actual annual generation may differ.

Electrical limitations

Final cable sizes, protection ratings, fault-current calculations and protection coordination require verification using actual installation conditions and equipment data.

Battery limitations

The battery is represented by a generic 48 V, 100 Ah LiFePO₄ specification. A real installation would require selection of a compatible battery model from the inverter manufacturer's approved/compatible battery options.

Utility limitations

Net metering and grid interconnection are shown conceptually. Actual utility approval and applicable Bangladesh requirements must be satisfied before implementation.

Installation limitations

No physical installation, testing, commissioning or operational verification was performed.

2.20 Conclusion

A complete conceptual hybrid rooftop solar PV system has been developed for an assumed residential installation in Rajshahi, Bangladesh.

The final design combines a **5.40 kWp PV array**, configured as **two independent 6-module series strings**, with a **5.0 kW hybrid inverter** and a **48 V, 100 Ah LiFePO₄ battery**.

The project goes beyond basic PV sizing by integrating load assessment, energy estimation, inverter compatibility, battery backup calculation, preliminary cable sizing, protection planning, physical CAD layouts and a complete electrical single-line diagram.

The resulting documentation demonstrates an end-to-end engineering workflow in which calculations support equipment selection and the selected equipment is then represented consistently in the physical layouts and SLD. Although the design is conceptual and requires site-specific verification before implementation, it provides a practical demonstration of solar PV system design, electrical drafting, documentation and engineering decision-making.

2.21 Appendix — Main Design Data

A. PV Module (JA Solar JAM54S40-450/LR)

- Rated power: 450 W
- V_{mp} : 32.82 V
- V_{oc} : 39.30 V
- I_{mp} : 13.71 A
- I_{sc} : 14.48 A
- Module quantity: 12

B. PV Array

- Total capacity: 5.40 kWp
- Configuration: 6S + 6S
- String capacity: 2.70 kWp
- Number of MPPTs used: 2

C. Inverter (GoodWe GW5000-ES-C10)

- Nominal AC output: 5.0 kW
- System: single-phase
- Nominal AC voltage: 230 V
- Frequency: 50 Hz
- MPPT inputs: 2
- Battery system: 48 V class

D. Battery

- Chemistry: LiFePO₄
- Nominal voltage: 48 V
- Capacity: 100 Ah
- Nominal energy: 4.8 kWh
- Assumed DoD: 80%

E. Preliminary Cabling

- PV cable: 4 mm² Cu
- AC cable: 6 mm² Cu
- Battery cable: final size to be selected according to actual battery/inverter requirements

F. Preliminary Protection

- DC isolators
- DC SPD as required
- Battery protection
- AC isolator
- Type II AC SPD
- 32 A, 2P AC MCB
- Main/backup DB protection
- PE/earthing and bonding